

Exercise 1.1: Safety vs. Security

Explain the difference between safety and security in the context of machine learning systems. Give one concrete example of a safety concern and one security concern for an ML-based autonomous vehicle.

Safety

- Protecting people and the environment from unintentional or accidental harm caused by the system.
- Comes from accidents, human error, unpredictable conditions or system that fails in its job.
- In ML perspective - this means a model that breaks down when the real world doesn't look like its training data. (Internal flaws)

Security

- Protecting the system from intentional attacks, unauthorised access.
- Threat comes from a person or group actively trying to exploit, manipulate, or gain unauthorised access to the system.
- In ML perspective - attackers specifically inputs design to fool model into making wrong predictions

Example

- Safety Concern - Shift at Night or in Snow
 - Models that were trained only on sunny daytime data, Weather and lighting conditions can change during a drive. When encountering these conditions, the model may fail. The model may fail to recognize something at night or in the snow. This is a safety concern, no attacker was involved
- Security concern - Attack on perception model
 - Attacker knowing the model can craft a designed perturbation to a stop sign, which caused the model to fail and detect the stop sign. This can cause for the vehicle to not follow the stop sign and it is a clear or deliberate act of the attacker to fail the model. So this is a security concern

Exercise 1.2: The EU AI Act

The EU AI Act classifies AI systems into risk categories.

- 1. Name the main risk categories and briefly describe each.**
- 2. Into which category would you place a perception system for a) an autonomous vehicle b) a ChatBot? Justify your answer.**
- 3. Research: What obligations does the Act place on providers of systems in that Category?**

Unacceptable Risk (Prohibited) — AI systems that pose a clear and fundamental threat to people's rights or safety are banned. Examples from slides include social scoring systems, real-time remote biometric identification in public spaces, and cognitive behavioural manipulation.

High Risk (Highly Regulated) — These systems are subject to stringent requirements due to their significant impact on critical areas such as public services, law enforcement, and safety components. Examples include AI applications in healthcare diagnostics, transportation management, and educational assessments. military and education as examples.

Limited Risk (Transparency Required) — The AI Act introduces specific disclosure obligations to ensure that humans are informed when necessary to preserve trust. For instance, when using AI systems such as chatbots, humans should be made aware that they are interacting with a machine so they can make an informed decision.

Minimal Risk (No Regulation) — Minimal risk is unregulated, including the majority of AI applications currently available on the EU single market, such as AI-enabled video games and spam filters. slides also mention general game AI as an example.

2. Classifying the Two Systems

a) Autonomous Vehicle Perception System → High Risk

- The AV perception system directly controls safety-critical decisions: whether to brake, decelerate, or continue driving.
- An AI system is considered high-risk if it is used as a safety component of a product covered by EU legislation — or if it is itself such a product.
- Transport is explicitly listed as a High Risk domain in the EU AI Act
- A failure in the perception model (e.g. missing a pedestrian) could directly cause death or serious injury

b) ChatBot → Limited Risk

- A general-purpose chatbot like ChatGPT does not make safety-critical autonomous decisions over the physical world.
- The AI Act introduces disclosure obligations — for instance, when using AI systems such as chatbots, humans should be made aware that they are interacting with a machine.
- This places standard chatbots in the **Limited Risk** category, requiring transparency but not the full compliance burden of High Risk.
- Note: if a chatbot were used in a high-risk context (e.g. medical diagnosis or legal decisions), its classification could be escalated.

3. Obligations for High-Risk AI Providers

Providers and deployers of high-risk AI systems will be subject to significant regulatory obligations, with enhanced thresholds of diligence, initial risk assessment, and transparency compared to AI systems not falling into this category. Specifically:

- **Risk Management System** — A risk management system is required for high-risk AI systems. The risk management process, which should span the entire life cycle of the AI system, must be iterative and updated as necessary.
- **Data Quality & Documentation** — Providers must draw up technical documentation, including training and testing processes and evaluation results.
- **Human Oversight** — Providers must ensure human oversight by developing mechanisms that allow human intervention in AI decision-making processes to prevent adverse outcomes.
- **Accuracy, Robustness & Cybersecurity** — Providers must design their high-risk AI system to achieve appropriate levels of accuracy, robustness, and cybersecurity.
- **Conformity Assessment & Registration** — Systems must be evaluated before deployment and registered with the relevant EU authority.
- **Incident Reporting** — High-risk AI system providers must report serious incidents when they occur — defined as malfunctions that lead to death or harm to a person's health, serious disruption to critical infrastructure, or serious harm to property or the environment.
- **Ongoing Monitoring** — Deployers must keep logs automatically generated by the high-risk AI system for a period appropriate to the intended purpose, of at least six months.

Exercise 1.3: Build the Right Model vs. Build the Model Right

Two complementary perspectives on ML quality (or software quality in general) can be summarized as:

- **“Build the right model”** — correctness: does the model solve the intended problem?
- **“Build the model right”** — robustness: is the model built and deployed with adequate care?

For a pedestrian-detection model in an autonomous vehicle, describe one failure captured by each perspective.

Build the Right Model -

Failure: The model detects the wrong visual feature

- The pedestrian detector achieves strong performance on the CARLA training set
- In a simulator like CARLA, pedestrians may consistently appear in specific contexts — for example, always on pavements next to kerbs, always rendered with the same lighting, or always accompanied by particular background elements.
- The model might learn to trigger on those contexts rather than on the actual shape of a human being.

- Here the model is solving the *wrong* problem. It was supposed to learn if there was a human but instead learned "are there pavement-adjacent pixels in this region?" The task specification itself was not correctly captured by the model.

Build the Model Right -

Failure: The model was trained only on sunny daytime data but deployed in rain and low light

- Models were "trained exclusively on sunny daytime data collected in the CARLA simulator", yet the operating environment acknowledges that weather and lighting conditions can change during a test drive
- Models will often give highly confident, wrong predictions on data too dissimilar from their training data. Under rainy or night conditions, the pedestrian detector may fail to fire — or fire with falsely high confidence — because the input distribution has shifted beyond what the model has ever seen.
- The underlying concept being learned (detect humans) is correct — the problem is that the engineering process was carried out inadequately. A sufficiently careful development process would have identified the ODD (Operational Design Domain) gap, tested under distribution shift, and either expanded the training data or constrained the deployment conditions to match the training data.

Exercise 1.4: First Impressions

Before looking at any data or model, think about the system described above.

1. Name three situations in which you would not trust this model's predictions. Explain why.

2. Which of the three output labels do you consider most safety-critical if missed?

Justify your choice.

3. What additional information would you want before deploying this model in a real vehicle?

1. Three Situations Where You Would Not Trust the Model

a) Weather — Rain, Fog, or Snow - The models were trained exclusively on clear/sunny weather in CARLA. Rain introduces visual noise, fog reduces visibility and blurs object boundaries, and snow can trick the shapes of pedestrians and vehicles entirely. models will often give highly confident, wrong predictions on data too dissimilar from their training data. The model has no learned representation for how a pedestrian looks through a snow filled lens.

b) Night-time - The training data was captured exclusively under daytime lighting. At night, the scene changes relying on artificial streetlights, headlight , and shadows rather than natural daylight.

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c) The Simulator-to-Real-World Gap

The training data was collected in a simulator, not from a real camera. CARLA renders synthetic scenes with perfect rendering- no lens flare, no compression artefacts, no dust on the lens, idealised textures, no motion blur from a real physical mount.

A real vehicle camera will produce images that differ in countless low-level ways from CARLA's output. Models trained entirely on simulator data may have quietly learned features that are specific to the rendering engine, and may fail when confronted with genuinely real-world images, even under sunny daytime conditions.

2. Most Safety-Critical Missed Label: Pedestrian

A missed pedestrian detection carries the highest potential for immediate, irreversible harm to a human being.

A missed vehicle detection could also cause a serious collision, but vehicles are larger and easier to detect, they are also better represented in urban environments.

A missed traffic light is serious at intersections.

3. Additional Information Needed Before Real Deployment

About the data and training:

ODD coverage analysis — What percentage of real-world driving scenarios are actually covered by the sunny CARLA training set? How many edge cases (night, rain, fog, occlusion, unusual clothing, cyclists) are entirely absent?

Class balance and dataset statistics — Are all three classes equally represented? A model trained on imbalanced data may achieve high overall accuracy while being nearly blind to the rare class (e.g. pedestrians appearing infrequently in simulator runs).

Target leakage check — Did the CARLA labelling process introduce any shortcuts, e.g. labels that correlate with metadata rather than actual visual content?

About model behaviour:

Per-class precision and recall, not just accuracy — As your lecture asks: "Is that [99% accuracy] good? What would random guessing give you?" Recall on the pedestrian class is the single most important metric for safety.

Calibration quality — "If the model predicts 'pedestrian' with 70% confidence, is this probability an accurate estimate?". Miscalibrated confidence means the planner cannot make informed threshold decisions.

Explainability evidence — Does the model attend to the actual pedestrian pixels, or to irrelevant background context?

Robustness to adversarial and OOD inputs — What happens to recall under FGSM perturbations and under rain/fog/night conditions?

About the deployment context:

Human operator fatigue profile — The course notes operators work 4-hour shifts with no auditory alerts. What is the realistic reaction time after prolonged monitoring? This directly affects whether the operator can compensate for a model failure in time.

System-level failure rate budget — What false negative rate on the pedestrian classifier is actually acceptable, given vehicle speed (up to 50 km/h), stopping distance, and operator reaction time? No deployment decision is meaningful without this figure.